Exercise Week 9

Advanced Applied Statistics

Fall 2025

Problem 1: Apparent Error vs. Cross-Validation Error*

Simulate a dataset of 100 observations with one predictor and one response variable, where

$$Y = 2X + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, 1).$$

Fit a linear model and compute the mean squared error (MSE) on the training data (apparent error). Then perform 5-fold cross-validation and compute the cross-validated MSE. Compare the two errors and reflect on why they differ.

Problem 2: Comparing Models Using Cross-Validation*

Simulate a dataset with 100 observations and 10 predictors:

$$Y = 3X_1 + 2X_2 - X_3 + \varepsilon,$$

where $\varepsilon \sim \mathcal{N}(0, 1)$, and $X_4, \ldots, X_{10}$ are noise predictors not involved in the true, generating model. Fit

(a) A full linear model using all predictors,

(b) A reduced model with only X_1, X_2, X_3.

Use 10-fold cross-validation to compare MSE for both models. Which model performs better? Why?

Problem 3: Understanding Covariance Penalty and Optimism*

Simulate a dataset with a moderate signal-to-noise ratio. Fit a model using least squares. Compute the apparent error (training MSE). Estimate the *optimism*, the gap between apparent error and expected training error, using:

$$\text{Optimism} \approx \frac{2\sigma^2 d}{n},$$

where d is the number of parameters. Add the optimism to the apparent error and compare it with the cross-validated error.

Problem 4: Mallows' C_p and AIC for Model Selection*

Use the `mtcars` dataset (R dataset, available also e.g. in pydataset in Python). Fit all possible subsets of predictors for predicting `mpg`. For each model, compute

- Mallows' C_p
- AIC

Plot C_p and AIC for all models. Which model would you choose based on C_p? AIC?

Problem 5: When AIC Fails in a High-Variance Case*

Simulate data with a small sample size (e.g., $n = 20$) and many predictors (e.g., $p = 15$). Fit several linear models and compute AIC. Also perform 10-fold cross-validation. Compare the model rankings from AIC and cross-validation. Reflect: When does AIC fail, and why?